

NexaMart Enterprise Data Warehouse — Milestone 2 Report

Group [N] · Members: Lead, M2, M3, M4, M5, M6 · Export to PDF as `nexamart_m2_report.pdf` for the ZIP. Authoritative definitions: `docs/glossary.md` . Verified counts: `.private/lead_cheatsheet.md` .

Executive Summary (Owner: Lead · ≤1 page)

The question. Was the Back-to-School campaign (8–28 Aug 2024) actually successful, or did every team just count something different?

The answer. On a certainty-segregated, anomaly-corrected basis it was **clearly positive** — campaign confirmed revenue ran ~**+50% above the pre-ramp run-rate** (+₹1.27 Cr EC incremental), not a failure and not the inflated **+34% Sales** celebrated. The gap between those two readings is not opinion; it is 24 resolved data anomalies. Sales counted **cancelled orders** (A1, ₹4.08M) and **seller-marked-sold listings** (A4, ₹1.72 Cr raw value) as confirmed revenue; a payment-after-cancel reversal liability (A2, flagged) and tax-basis mismatches (A3) distorted the rest. *(The briefs "+11%" reference figure is scenario framing — see S5; this warehouse does not reproduce it as a growth rate.)*

How we got one number. We resolved every anomaly in corrected Silver (audit trail preserved, nothing deleted), rebuilt the affected Gold facts, validated with 10 checks, and exposed certainty-segregated KPI views in `NEXAMART_MARTS` . Confirmed revenue and **Estimated Classified GMV** (NexaLocal offline, ±35% band) are reported **separately and never added** — the single most common way the legacy teams double-counted.

The headline numbers (GSV → NCR waterfall in S4; certainty-labelled metrics in S5): GSV **₹141.5 Cr** minus cancelled EC revenue (₹4.08M) and POS line discounts (₹2.75 Cr) = **NCR ₹138.35 Cr**, the one trusted top-line. Return refunds (₹0.65 Cr) and **Estimated Classified GMV** (₹1.26 Cr point, ±35% band, ESTIMATED) are held **separately and never added in**. On the campaign window, EC confirmed revenue is **+₹1.27 Cr incremental** (B1 session-bridge attribution, INFERRED).

Section 1 — Anomaly Resolution Report (All members; owner per anomaly)

For **each Category A** anomaly: detection method, root cause, the specific PySpark fix logic, the post-fix verification result (**affected row count before → after**), and the business impact in financial/operational terms. For **each Category B**: precise description of the ambiguity, the chosen interpretation, the PySpark implementation, the business justification for choosing it over alternatives, **and the quantified impact had the alternative been chosen** (brief §8.2).

Cross-references: detection = `sql/anomaly_discovery.sql` ; resolution code = `notebooks/05_anomaly_resolution.ipynb` ; per-anomaly docs = `sql/anomaly_resolution.sql` .

Detection verified (run 14 Jun 2026 as `NEXAMART_ENGINEER` against live `NEXAMART_SILVER`). 18 of 24 anomalies matched the expected fingerprints EXACTLY: A1=94, A2=27, A4=449, A5=5, A6=8, A7=10, A8=21, A9=1, A10=12, A11=178, A12=3, A13=3 rings, A14=18 strict, A15=25, A16=7, B1=126, B3=175, B7=25. The remaining six are, by design, **not single-count SQL detections**: A3 (schema-wide tax-basis comparison), B4 (catalogue-match threshold), B6 (Estimated-GMV signal model), and B5/B8 (probabilistic identity-merge / seller-trust scores) are computed in the `05` PySpark resolution; B2 (cross-period partial refund, 1 row) is finalised there too. Schema notes baked into the predicates: member-built Silver tables use base column names (`created_at` , `snapshot_date`), not the lead-built `_parsed` variants; `captured_ts` is a `TIMESTAMP` while `status_ts_parsed` is epoch-NUMBER; A9 is the cross-source `silver_ts_seller_listings` catalogue mismatch.

Category A

ID	Detection (live Silver predicate)	Root cause	Fix (method)	Before → After	Business impact	Owner
A1	<code>order_status='CANCELLED' AND subtotal_excl_tax>0</code> ✓94	cancelled orders left revenue on the line	ZEROED_CANCELLED_REVENUE	94 → 0	₹4.08M (₹0.41 Cr) over-count removed	M5
A2	captured PG txn (captured_ts not null) after earliest CANCELLED status	capture fired post-cancel	FLAGGED_REVERSAL_REQUIRED	27 → 0	27 captured txns held as reversal liability (excl. NCR)	M5
A3	POS incl-tax vs EC excl-tax basis differ	mixed tax bases summed naively	NORMALISED_TAX_EXCLUSIVE	schema-wide	₹16.25 Cr comparable basis	M5
A4	<code>event_type_code='SELLER_SOLD'</code> ✓449	seller-sold counted as confirmed revenue	RELABELLED_ESTIMATED_GMV	449 → 0 confirmed	₹1.72 Cr moved to ESTIMATED	M6
A5	<code>atp_qty>0 AND physical_qty=0</code> ✓5	ATP not decremented to physical	CORRECTED_ATP_TO_ZERO	5 → 0	oversell risk removed	Lead
A6	<code>sellable_qty<0 OR physical_qty<0</code> ✓8	negative inventory written	CORRECTED_ATP_TO_ZERO	8 → 0	stockout analysis fixed	M4
A7	<code>inspection_status='PENDING' AND restocked_qty>0</code> ✓10	restock before inspection	ZEROED_RESTOCK_PRE_INSPECTION	10 → 0	ATP inflation removed	Lead
A8	stores 3/7/12 × (1–7 Aug) absent from snapshots	snapshots not captured in ramp-up	RECONSTRUCTED_SNAPSHOT	~21 reconstructed	allocation accuracy (1–7 Aug)	Lead
A9	<code>silver_ts_seller_listings</code> : listing 42 / NX-TECH-0001 catalogue mismatch	seller feed contradicts catalogue	APPLIED_CANONICAL_PRODUCT	1 → 0	category revenue corrected	M3
A10	<code>condition_on_receipt='OPENED' AND restocked_as_condition='NEW'</code>	open-box restocked as NEW	CORRECTED_CONDITION_OPEN_BOX	12 → 0	price premium quantified	M4
A11	<code>customer_id='9999'</code> ✓178	guest placeholder collides w/ real loyalty	REKEYED_GUEST_BUCKET	178 → 0	customer profile de-collided	M2
A12	self-join <code>image_hash</code> +seller, prior SOLD/EXPIRED, ACTIVE relist	listing relisted after sold	LINKED_RELISTING_EXCLUDED	3 pairs	GMV double-count removed	Lead
A13	<code>image_hash</code> shared by ≥2 seller accounts ✓3 rings	coordinated listing ring	FLAGGED_FRAUD_RING	rings flagged	trust & safety	M6

ID	Detection (live Silver predicate)	Root cause	Fix (method)	Before → After	Business impact	Owner
A14	<code>delivered_ts < pickup_ts</code> (strict) ✓18	courier clock drift	CORRECTED_DELIVERY_TS / ESCALATED_MANUAL_REVIEW	18 (>72h escalated)	on-time rate corrected	Lead
A15	<code>days_post_delivery < 0</code>	review before delivery	SET_VERIFIED_PURCHASE_FALSE	25 → 0 verified	review-score KPI corrected	M6
A16	<code>is_duplicate_flag=TRUE</code> ✓7	same complaint via many channels	DEDUPED_CANONICAL_CASE	7 → deduped	complaint volume corrected	M6

Category B

ID	Ambiguity	Chosen interpretation	Quantified alternative	Owner
B1	promo-less in-window + prior UTM	attribute 102 (conf 0.85)	not attributing understates ~₹0.94 Cr confirmed	Lead
B2	Aug sale / Sep refund	recognise in return period	original-period reduces Aug NCR ~₹2.16 L	Lead
B3	175 PCK null-ref movements	probable missing-ref (INFERRED)	shrinkage reading overstates loss	Lead
B4	NL title → catalogue	≥0.75 match / 0.65–0.75 review	lower threshold injects false matches	M3
B5	cross-channel identity	merge ≥0.90	0.70 over-merges distinct customers	M2
B6	Estimated GMV formula	weighted signals + ±35% band	seller-sold-only mis-states	M2
B7	BOPIS no pickup event	treat as fulfilled (scan miss)	not-collected understates fulfilment	Lead
B8	seller trust score	weighted 8-signal → 5 tiers	equal-weight unacceptable	M2

Per-anomaly detail (detection → root cause → fix → verification → impact)

Resolution code: `notebooks/05_anomaly_resolution.ipynb` (one cell per anomaly); post-resolution verify: `sql/anomaly_resolution.sql`. Every corrected row keeps `anomaly_flag=TRUE` + its original `anomaly_reason_code` and gains `resolution_applied=TRUE` + `resolution_method` (+ `b_classification` for Category B) — nothing is deleted (brief §8.1).

- A1 — Cancelled orders carrying revenue.** 94 EC orders had `order_status='CANCELLED'` but a positive `subtotal_excl_tax`; the cancellation workflow set the status without reversing the revenue line. Fix: flag `CANCELLED_WITH_REVENUE`, zero the revenue (`ZEROED_CANCELLED_REVENUE`). In Gold the keystone fact keeps the gross line value (for GSV) but sets `confirmed_revenue_excl_tax=0` for cancelled lines, so NCR excludes them. Verify → 0 unresolved. **Removes the ₹4.08M (₹0.41 Cr) cancelled-revenue over-count that inflated Sales' +34% claim** (`vw_revenue_leakage` CANCELLATION = ₹4,078,605).
- A2 — Payment captured after cancellation.** 27 captured PG transactions (26 strictly after the cancellation timestamp) sit against cancelled orders. Fix: `FLAGGED_REVERSAL_REQUIRED` (kept, excluded from NCR) — held as a reversal liability (the captured-payment exposure sits in the PG source, outside the Gold facts), never counted as revenue.
- A3 — Mixed tax bases.** POS stores tax-inclusive amounts, EC tax-exclusive; naively summing them is meaningless. Fix: derive a tax-exclusive POS measure (`NORMALISED_TAX_EXCLUSIVE`) so every channel reconciles on one basis. Reconciles the ~₹16.25 Cr naive cross-channel figure.
- A4 — Seller-marked-sold counted as revenue.** 449 `SELLER_SOLD` NexaLocal events were summed as confirmed GMV. Fix: relabel certainty `ESTIMATED` (`RELABELLED_ESTIMATED_GMV`); they feed the B6 model at weight 0.60, never Confirmed GMV. Moves ₹1.72 Cr out of confirmed totals.

- **A5 / A6 — Impossible inventory.** 5 warehouse SKU-dates showed ATP>0 with physical=0; 8 store snapshots carried negative quantities. Both corrected to 0 (`CORRECTED_ATP_TO_ZERO` , A6 also sets an oversell flag), removing oversell risk and fixing stockout analysis.
- **A7 / A10 — Return-receipt handling.** 10 receipts restocked while `inspection_status='PENDING'` (`ZEROED_RESTOCK_PRE_INSPECTION` , ATP inflation removed); 12 open-box returns mislabelled NEW (`CORRECTED_CONDITION_OPEN_BOX` , price-premium quantified). Distinct predicates, overlapping sets.
- **A8 — Missing snapshot days.** Stores 3/7/12 each missing 7 snapshot days (1–7 Aug, pre-ramp). The reconstructed sibling `silver_store_inventory_snapshots_reconstructed` (last-known + interpolation) is resolution-stamped `RECONSTRUCTED_SNAPSHOT` , certainty INFERRED — never CONFIRMED, never campaign-window.
- **A9 — SKU/product mismatch.** Listing 42 cites SKU `NX-TECH-0001` (a laptop) but describes a phone case. Fix: catalogue wins (`APPLIED_CANONICAL_PRODUCT`), 1 → 0.
- **A11 — Placeholder-ID collision.** 178 guest EC orders share `customer_id='9999'` with a real dormant loyalty account (Sarah Chen). Fix: rekey guests to `GUEST-{session_id}` (`REKEYED_GUEST_BUCKET`), de-colliding the real customer's profile.
- **A12 / A13 — Classified integrity.** 3 relisted-after-sold pairs (same seller + image hash) linked and excluded from GMV (`LINKED_RELISTING_EXCLUDED` , reliability LOW); 3 image-hash rings spanning ≥2 seller accounts (18 listings) flagged `FLAGGED_FRAUD_RING` .
- **A14 — Courier clock drift.** 18 shipments show a DELIVERED event before PICKED_UP. For $|\text{delta}| \leq 72\text{h}$ the delivered timestamp is corrected to pickup + 36h (`CORRECTED_DELIVERY_TS`); > 72h is escalated, not auto-corrected (`ESCALATED_MANUAL_REVIEW`) — the expected residual after resolution.
- **A15 / A16 — CX signals.** 25 reviews posted before delivery set `is_verified_purchase=FALSE` (`SET_VERIFIED_PURCHASE_FALSE`); 7 duplicate support cases resolved to their canonical key (`DEDUPED_CANONICAL_CASE`) so complaint volume counts distinct incidents.
- **B1 — Attribution bridge.** Of 126 promo-less in-window orders, 102 had a prior `BTS2024` UTM session within the 2h bridge → `ATTRIBUTED` at confidence 0.85. **Alternative:** not attributing understates campaign revenue by ~₹0.94 Cr confirmed (₹9,432,000 over the 102 attributed orders; ₹1.02 Cr gross).
- **B2 — Partial-refund period.** 1 cross-month partial refund recognised in the **return** period (`RECOGNISE_IN_RETURN_PERIOD`); both period impacts are exposed. Original-period recognition would cut Aug NCR by ~₹2.16 L.
- **B3 — Null-reference movements.** 175 NULL- `reference_number` picks (all PCK) classified `PROBABLE_MISSING_REF` (INFERRED, back-fill lag) rather than treated as shrinkage, which would overstate loss.
- **B4 — Catalogue matching.** NexaLocal free-text titles matched to the catalogue with rapidfuzz: ≥0.75 MATCHED, 0.65–0.75 MANUAL_REVIEW, <0.65 UNMATCHED. A lower threshold injects mis-linked demand signals.
- **B5 — Identity merge.** A single 0.92-confidence cross-channel pair merged (`MERGED` , threshold ≥0.90); a 0.70 threshold would over-merge distinct customers.
- **B6 — Estimated Classified GMV.** Locked weights (SELLER_SOLD 0.60 / PHN_REVEAL 0.15 / CHAT 0.08 / OFFER_ACC 0.30) × listing confidence, ±35% band, labelled ESTIMATED — surfaced only via `vw_estimated_classified_gmv` , never summed into confirmed revenue.
- **B7 — BOPIS scan miss.** 25 BOPIS orders marked DELIVERED with no `BOPIS_COLLECTED` scan treated as fulfilled (`TREAT_AS_FULFILLED` , ~13% scan-miss baseline) + `collection_unconfirmed=TRUE` , excluded from the BOPIS SLA.
- **B8 — Seller trust.** Flagged high-risk sellers escalated to `UNDER_REVIEW` via a weighted 8-signal composite (not equal-weight) → 5 tiers; UNDER_REVIEW is reversible (30-day monitoring), unlike suspension.

Section 2 — Gold Rebuild Summary (Owner: Lead, with per-fact owners)

Corrected Silver flows to Gold via **partial rebuilds** (`notebooks/06_gold_rebuild.ipynb`) — only the facts an affected Silver entity feeds are re-pointed at corrected Silver and overwritten; the 8 static lookup dims (`dim_date` , `dim_channel` ,

`dim_step`, `dim_payment_method`, `dim_delivery_method`, `dim_listing_condition`, `dim_return_reason`, `dim_promotion`) are **rebuild-exempt** (no Silver feed).

Facts rebuilt (9): `fact_ecommerce_order_line` (A1/A2/A3/A11/B1 — keystone, with the GSV/NCR split), `fact_order_fulfilment` (A14/B7), `fact_store_inventory_snapshot` (A6/A8), `fact_warehouse_inventory_snapshot` (A5), `fact_inventory_transaction` (B3), `fact_classified_listing_snapshot / _event` (A4/A12/A13/B6), `fact_customer_review` (A15), `fact_customer_complaint` (A16). Each rebuild carries an inline `assert_grain` (`COUNT(*) = COUNT(DISTINCT grain)`).

Row counts: corrections change measures and flags, **not** grain, so before/after row counts are **identical** per table — the notebook's final cell prints the before/after delta as proof (a non-zero delta is treated as a defect). Descriptive dims (`dim_customer` / `dim_product` / `dim_seller`) are left intact: facts carry their dim FK as `surrogate_key(natural_code)`, so every fact→dim join still resolves, and the SCD2 dim refresh (carrying the A9/B5/B8 descriptive corrections) is a member follow-up. `fact_store_sale_line` / `fact_return_line` / `fact_seller_performance_snapshot` have no in-repo M1 build logic and are noted here rather than rebuilt blind.

Before → after row counts (Phase-B run, 15 Jun 2026 — every delta 0, confirming grain-preserving corrections):

Rebuilt fact	Before	After	Δ
<code>fact_ecommerce_order_line</code>	1,840	1,840	0
<code>fact_order_fulfilment</code>	963	963	0
<code>fact_store_inventory_snapshot</code>	217,800	217,800	0
<code>fact_warehouse_inventory_snapshot</code>	38,610	38,610	0
<code>fact_inventory_transaction</code>	468,455	468,455	0
<code>fact_classified_listing_snapshot</code>	1,253	1,253	0
<code>fact_classified_listing_event</code>	38,706	38,706	0
<code>fact_customer_review</code>	377	377	0
<code>fact_customer_complaint</code>	129	129	0

All nine rebuilt facts pass their inline `assert_grain` (`COUNT(*) = COUNT(DISTINCT grain)`), and a second full Run-all reproduces these counts identically (idempotent). The zero deltas are the correctness proof: A1/A2/A4/A14/B1/B3 etc. re-value measures and set flags on existing rows — no row is added or deleted.

Section 3 — Validation Outcomes (Owner: Lead)

`sql/validation_suite.sql` runs 10 checks against `NEXAMART_GOLD` after every rebuild; each returns the offending rows (target 0, except Check 7 which expects ≥ 1 row per fact in the campaign window).

#	Check	What it proves
1	Completeness	all 27 Gold tables non-empty (UNION-ALL row counts)
2	Referential integrity	no orphan <code>date_key</code> (computed identically everywhere) across the facts
3	Grain	<code>COUNT(*) = COUNT(DISTINCT grain)</code> per declared grain
4	Additive sanity	<code>net = gross - discount</code> on <code>fact_store_sale_line</code> (no return column at this grain; returns live in <code>fact_return_line</code>)
5	Semi-additive guard	no MARTS view <code>SUM</code> s ATP across dates (<code>vw_atp_sku_loc_date</code> exposes the grain, no <code>SUM</code>)
6	Certainty completeness	no NULL <code>metric_certainty_level</code> in any of the 14 facts

#	Check	What it proves
7	Campaign coverage	≥1 row per campaign-accepting fact in 8–28 Aug (via <code>dim_date</code> join)
8	Inventory reconciliation	snapshot $\Delta(\text{physical_qty}) = \text{net signed movement, per store} \times \text{product}$
9	Certainty segregation	no Finance view row is ESTIMATED while <code>is_confirmed_transaction <> FALSE</code> ; the estimated view is 100% ESTIMATED
10	Temporal consistency	no auto-correctable (≤72h) delivered-before-picked remains post-A14 (>72h rows flagged UNRELIABLE)

Iteration log (from the Phase-B runs, 15 Jun 2026 — honest failure reporting is graded above an implausible first-run pass):

Iteration	Checks failed	Root cause	Fix applied	Re-run result
1	Check 4 (additive sanity); Check 8 (inventory reconciliation)	C4 referenced <code>return_amount</code> , which does not exist on the member-built <code>fact_store_sale_line</code> (real measures: <code>gross_amount</code> , <code>discount_amount</code> , <code>net_amount</code> , <code>tax_amount</code> , <code>cogs_amount</code>) → invalid-identifier error. C8 moves CTE summed all-time store movements against a campaign-window snapshot delta — incomparable by construction (1,095 offending pairs).	C4 → assert the identity that actually holds at this grain, <code>net_amount = gross_amount - discount_amount</code> . C8 → window moves to 8–28 Aug to match the snapshot bounds (<code>quantity_delta</code> is already signed: PICK/DMG –, RCVD/RET +).	C4 = 0 broken of 24,507 (PASS). C8 = 863 of 1,100 store×product pairs still residual (237 reconcile exactly).
2 (final)	none blocking	C8 residual root-caused: store inventory snapshots are independent periodic physical counts , while <code>fact_inventory_transaction</code> records only PICK/RCVD/RET/DMG — store POS sell-through is not in the movement ledger , so the two facts cannot reconcile to unit tolerance (avg residual 7.4 units; 515 pairs within ±2). This is a data-sourcing observation, not an anomaly-resolution defect; tolerance 0 is left deliberately strict so the check keeps surfacing the gap.	Documented (no code change).	9 of 10 checks PASS ; Check 8 = documented reconciliation residual.

Final state per check (live NEXAMART_GOLD / NEXAMART_MARTS , run 15 Jun 2026): C1 27/27 tables non-empty (min row_count 2); C2 0 orphan `date_key` ; C3 0 duplicate-grain groups; C4 0 broken of 24,507; C5 0 ATP-summing views; C6 0 NULL `metric_certainty_level` ; C7 every campaign-accepting fact ≥ 1 in window (min 227); C8 documented residual (above); C9 0 certainty-segregation breaches; C10 0 auto-correctable delivered-before-picked remaining.

Section 4 — KPI Reconciliation Report (Owner: Lead)

The seven teams disagreed because each summed a **different measure on a different basis**. One trusted number — **Net Confirmed Revenue (NCR)** — is reached from Gross Sale Value (GSV) by a named, quantified waterfall (`NEXAMART_MARTS.vw_gsv` , `vw_ncr` , `vw_revenue_leakage` ; method in `docs/reconciliation_method.md`).

GSV → NCR waterfall (real figures, deployed views, run 15 Jun 2026)

The two GSV→NCR deductions that the deployed `vw_gsv` / `vw_ncr` actually quantify are A1 cancelled EC revenue and POS line-level discounts (store NCR = `net_amount` = `gross_amount` - `discount_amount`). A2 (payment-after-cancel) carries no extra ₹ — those payments sit on orders already zeroed by A1, and are held as a reversal *liability*, not revenue; A3 normalises POS to a tax-exclusive basis (a basis correction, not a ₹ line); B2's single partial refund (₹2.16 L) is recognised in the return period. All figures below are across the four in-scope phases (1 Jan – 14 Sep 2024).

Gross Sale Value (GSV, tax-exclusive, all confirmed channels)	₹141.51 Cr	(₹1,415,087,895)
- Cancelled EC revenue	← A1 ZEROED_CANCELLED_REVENUE	₹0.41 Cr (₹4,078,605; 94 orders / 157 lines)
- POS line-level discounts	← gross→net at the store grain	₹2.75 Cr (₹27,502,800)
= Net Confirmed Revenue (NCR)	₹138.35 Cr	(₹1,383,506,490) ← the trusted top-line

Two leakages/measures are surfaced **separately**, never folded into NCR: platform **return refunds**

(vw_revenue_leakage , RETURN_REFUND) = **₹0.65 Cr** (₹6,450,710, from fact_return_line.refund_amount), and

Estimated Classified GMV (NexaLocal offline, vw_estimated_classified_gmv , ESTIMATED, ±35% band) = **₹1.26 Cr point** (₹12,603,544; band ₹0.82 Cr – ₹1.70 Cr). The latter is the B6-weighted estimate (SELLER_SOLD 0.60 / PHN_REVEAL 0.15 / CHAT 0.08 / OFFER_ACC 0.30 × listing asking_price); the **raw** seller-marked-sold value that A4 pulled out of *confirmed* GMV is ₹1.72 Cr (SUM(asking_price) of the 449 SELLER_SOLD listings). Confirmed GMV (vw_confirmed_gmv , = NCR by construction) and Estimated Classified GMV are **reported separately and never added** — the single most common legacy double-count.

Data note: cogs_amount is unpopulated in the member-built sale facts, so vw_gross_margin_by_channel / vw_net_margin_after_fulfilment currently return margin = revenue (numerator = denominator); margin % is therefore not yet meaningful and is flagged for the SCD/COGS member follow-up rather than reported as a KPI.

Why each team's number differed — claimed vs warehouse-reproducible

Each team's headline came from summing a different measure on a different basis. The brief's claimed growth rates (Sales +34%, Finance +11%, ...) are **not reproducible as growth rates** from this warehouse; what the deployed views do reproduce is the ₹ figure each team mis-stated and the corrected one beside it. Every figure below traces to a live NEXAMART_MARTS view (run 15 Jun 2026) and carries its certainty.

Team	Claimed (brief)	Warehouse-reproducible figure (certainty)	Anomaly(ies)
Sales	+34% revenue	confirmed revenue overstated by ₹2.13 Cr : –₹0.41 Cr cancelled EC orders removed (vw_revenue_leakage , CONFIRMED) + ₹1.72 Cr seller-marked-sold reclassified out of confirmed to ESTIMATED	A1, A4/B6
Finance	+11% (reference)	NCR ₹138.35 Cr — the trusted basis every other number reconciles to (CONFIRMED)	—
Marketplace	inflated GMV	Estimated Classified GMV ₹1.26 Cr B6-weighted vs ₹1.72 Cr raw (–₹0.46 Cr over-weight; ESTIMATED) + duplicate-listing inflation 1.015× (19 of 1,253 listings; CONFIRMED)	B6, A12
Inventory	"only 3 stockouts"	5 SKU-location-days at ATP = 0 and 5 residual oversell-risk snapshots, all inside the campaign window ; inventory accuracy 99.47% (216,645/217,800; CONFIRMED)	A5, A6
Ecommerce	abandonment +22%	cart abandonment 41.5% (684/1,647), payment-failure 7.0% (67/963), on-time delivery only 54.2% (340/627) — abandonment tracks supply/delivery failure, not demand (CONFIRMED, whole-period)	A5/A6, A14
Store Ops	pickup handling ×3	896 BOPIS orders made visible (avg readiness 16.9 h); BORIS = 0 in window (CONFIRMED)	B7, BORIS split
Support	340 complaints	A16 cross-channel case de-duplication applied in S1; complaint volume is not surfaced as a MARTS KPI (no support view)	A16
Cross-cutting	—	tax-basis (A3) normalises POS to tax-exclusive; refund-period (B2) moves the ₹2.16 L partial refund to the return period — both make cross-channel sums comparable	A3, B2

Grain note: the funnel/conversion and seller figures above are **whole-period** (those views are not phase-split); the phase-split numbers (GSV / NCR / Confirmed GMV by Baseline | Campaign | Post) are in the waterfall above and in S5.

On the campaign window itself (8–28 Aug, 21 days), vw_campaign_incremental_revenue puts EC confirmed campaign revenue at **₹1.79 Cr** against a length-normalised baseline of **₹0.52 Cr** → **+₹1.27 Cr incremental (CONFIRMED)**; B1's session-bridge attribution (102 of the 126 promo-less in-window orders, confidence 0.85) is the INFERRED mechanism behind that lift. The data-backed campaign verdict is in S5.

Section 5 — Campaign Performance Conclusion (Owner: Lead)

Position. On a **certainty-segregated, anomaly-corrected basis** the **Back-to-School** campaign was clearly **positive**. Campaign-window confirmed revenue ran at **₹11.69M/day versus the ₹7.78M/day pre-ramp run-rate (+50%)** — and well above the longer baseline — with **+₹1.27 Cr EC incremental** confirmed revenue (vw_campaign_incremental_revenue : campaign ₹1.79 Cr vs a length-normalised baseline ₹0.52 Cr). The legacy **+34% Sales** headline was inflated by counting two things that are **not** confirmed revenue: cancelled EC orders (A1, ₹4.08M, now zeroed) and seller-marked-sold classified listings (A4, ₹1.72 Cr raw value, now relabelled ESTIMATED and reported only in the ±35% band — B6 point estimate ₹1.26 Cr). Removing that inflation is not opinion; it is 24 resolved anomalies. *(The brief's specific "+11% vs +34%" reference figures are scenario framing and are **not reproducible as growth rates** from this warehouse; the defensible, data-backed verdict is the clean positive uplift above, with the inflation explicitly segregated rather than a single contested percentage.)* Operational caveats — on-time delivery at 54% (vw_on_time_delivery_rate , 340/627) and the inventory snapshot-vs-ledger reconciliation residual (S3) — are conditions the campaign **exposed**, not caused.

Each conclusion metric is reported **with its certainty level, never conflated** — ESTIMATED values are never summed into CONFIRMED totals:

Conclusion metric	Source view	Certainty	Reading (live, 15 Jun 2026)
Net Confirmed Revenue	vw_ncr , vw_campaign_incremental_revenue	CONFIRMED	₹138.35 Cr total; campaign ₹24.55 Cr ; + ₹1.27 Cr EC incremental
Confirmed GMV	vw_confirmed_gmv	CONFIRMED	₹138.35 Cr (= NCR by construction); campaign-phase ₹24.55 Cr
Estimated Classified GMV (±35%)	vw_estimated_classified_gmv	ESTIMATED	₹1.26 Cr point (band ₹0.82–1.70 Cr); reported separately, never added to NCR
Net margin after fulfilment	vw_net_margin_after_fulfilment	CONFIRMED	deferred — cogs_amount unpopulated in source facts; margin % not yet meaningful (member COGS follow-up)
Inventory health (stockout / accuracy)	vw_stockout_rate , vw_inventory_accuracy_rate	CONFIRMED	accuracy 216,645/217,800 = 99.5% ; stockouts ≤ 1/195 SKU-day on only a few campaign days (A5/A6 corrected)
Fulfilment (on-time, BOPIS)	vw_on_time_delivery_rate , vw_bopis_pickup_readiness_time	CONFIRMED	on-time 340/627 = 54% ; BOPIS readiness 16.9h avg over 896 orders (A14 corrected; >72h escalated)
Customer satisfaction	vw_validated_report_rate , review/complaint views	CONFIRMED / INFERRED	validated-report 18/19 ; A15 (25 reviews) set unverified, A16 (7 cases) deduped
Classified engagement	vw_listing_contact_rate , vw_active_listing_count	CONFIRMED	538 active listings, 564 contacts; relisting 3/1,253; A12/A13 fraud rings excluded

Certainty discipline holds end-to-end: validation Check 9 confirms 0 ESTIMATED rows leak into any confirmed Finance view and vw_estimated_classified_gmv is 100% ESTIMATED; Check 5 confirms no view sums ATP across dates. The ₹1.26 Cr estimated classified GMV is **never** added to the ₹138.35 Cr NCR — the two sit in different certainty bands by design.